Ex: 13.2 to binary: (W2.1)

integer:

13/2 : 6 Remainder 1 LSB
6/2 : 3 " 0
3/2 : 1 " 1
1/2 : 0 " 1 MSB

1101

fraction:

0.2 × 2 = 0.4 MSB
0.4 × 2 = 0.8
0.8 × 2 = 1.6
0.6 × 2 = 1.2
0.2 × 2 = 0.4 LSB

.00110011...

The result: 1101.00110011 ~.

In octal:

1101. 001 100 110 011 001
1 5 . 1 4 6 3 1

$(15.14631463...)_8$

In Hexadecimal:

1101. 0011 0011 0011 001!
D . 3 3 3
(13)

$(D.33...)_{16}$

Most significant 6.7: The change in this bit has maximum effect on the magnitude of number. (for integers: MSB is the left most, for fractions.

Ex: a ⟶ 9

Least significant bit: The change in this bit has minimum effect on the magnitude of the number (if it is at right most)

* FUNDAMENTAL MATH. OPS.

* BINARY ADDITION

```
  1 1 1 1 1 1 ←
    1 1 1 1 0 1  → 61
+   1 0 1 1 1 1  → 23
-------------------
  1 0 1 0 1 0 0  → 84
```

2 ≥ Base ⇒ Subtract a base
2 − Base = 0
(1)

1+1+1 = (2) + 1 = 0+1 = 1

↓ 0

1

* BINARY SUBSTRACTION

Borrow a base

```
  1 0 0 1 0 1  → 77
    1 0 1 1 1  → 23
-------------------
  0 1 1 0 1 1 0  → 54
```

* BINARY MULTIPLICATION:

```
    1 0 1 1 1  → 23
x   1 0 1 0  → 10
-------------
    0 0 0 0 0
  1 0 1 1 1
0 0 0 0 0
+ 1 0 1 1 1
-------------
  1 1 1 0 0 1 1 0  → 230
```

* COMPLEMENTS

* There are 2 types of complements for each base-B system: Radix complement and diminished radix comp.

* DIMINISHED RADIX COMPLEMENT: (r-1)s complement

	Ex $(2368)_{10}$
The Number: N	
base : r	10
Number of digits: n	4
(r-1)'s complement: $(r^n-1)-N$	$(10^4-1)-2368$ } 9's complement $=7631$

Example for 7 bit binary numbers: #

1's complement: $(1011000)_2$

$(0100111)_2$ → this is 1's complement.

* Diminished radix complement can be computed digit by digit because subtraction from (r^n-1) will never require a borrow.

* If you add a number and its diminished radix complement, the result is the maximum number in the base system within the number of digits.

* RADIX COMPLEMENT

The r's complement of an n digit number in base r is defined as $r^n - N$ for $N \neq 0$ and as 0 for $N = 0$. Basically it can be obtained by adding 1 to the (r-1)'s complement.

The number: N, base: r, number of digits: n

$\Rightarrow$ r's complement: $r^n - N$ = (r-1)'s complement + 1

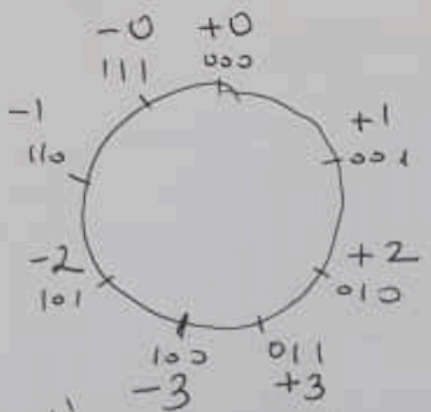
Ex: The 10's compl. of $(012398)_{10} \rightarrow (987612)_{10}$

$(246700)_{10} \rightarrow (753300)_{10}$

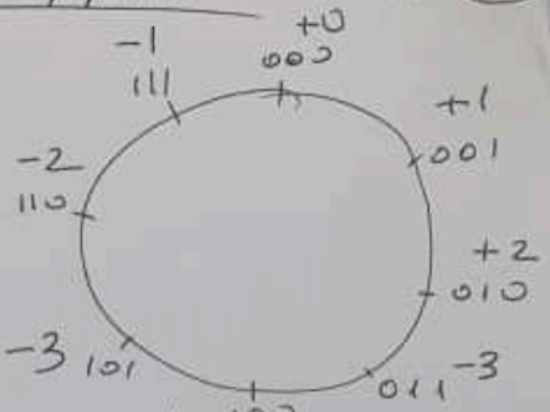
The 2's complement of $(1101100)_2 \rightarrow (0010100)_2$

$(0110111)_2 \rightarrow (1001001)_2$

Sign – Magnitude Representation 5



1's complement



2's complement

We can represent one more neg. numbe using 2's complement representation.

The Math. ops. using r's complements:

* Substraction: $(M-N)$

1) Add M and the r's complement of N.

$$(M + r^n - N) = \text{Result}.$$

2) If $M \geq N \Rightarrow$ the sum will produce an end carry r^n. Discard the end carry

3) if $M < N \Rightarrow$ the sum doesn't produce an end carry but it is the r's complement of $(N-M)$. $\{r^n - (N-M)\}$. Take the result's r's complement again and place a negative sign in front.

Ex:

$$\begin{array}{r} 9 \\ -5 \\ \hline 4 \end{array} \quad \begin{array}{l} 9 \\ 5 \\ + \\ \hline 14 \end{array}$$

10's comp → 5

14 ⇒ the result is +4

$$\begin{array}{r} 5 \\ -9 \\ \hline -1 \end{array} \quad \begin{array}{l} 5 \\ 1 \\ + \\ \hline 6 \end{array}$$

6 → No end carry.

10's complement

4 ⇒ the result is −4.

in binary :

$$\begin{array}{r} 2 \\ -3 \\ \hline -1 \end{array} \quad \begin{array}{l} 2 \to 010 \\ -011 \\ \hline \end{array} \quad \begin{array}{l} 010 \\ 101 \\ + \\ \hline 111 \end{array}$$

2's comp

(2's complement)

−001

Over flow cases

$$\begin{array}{r} -2 \\ -3 \\ \hline 1 \end{array}$$

overflow